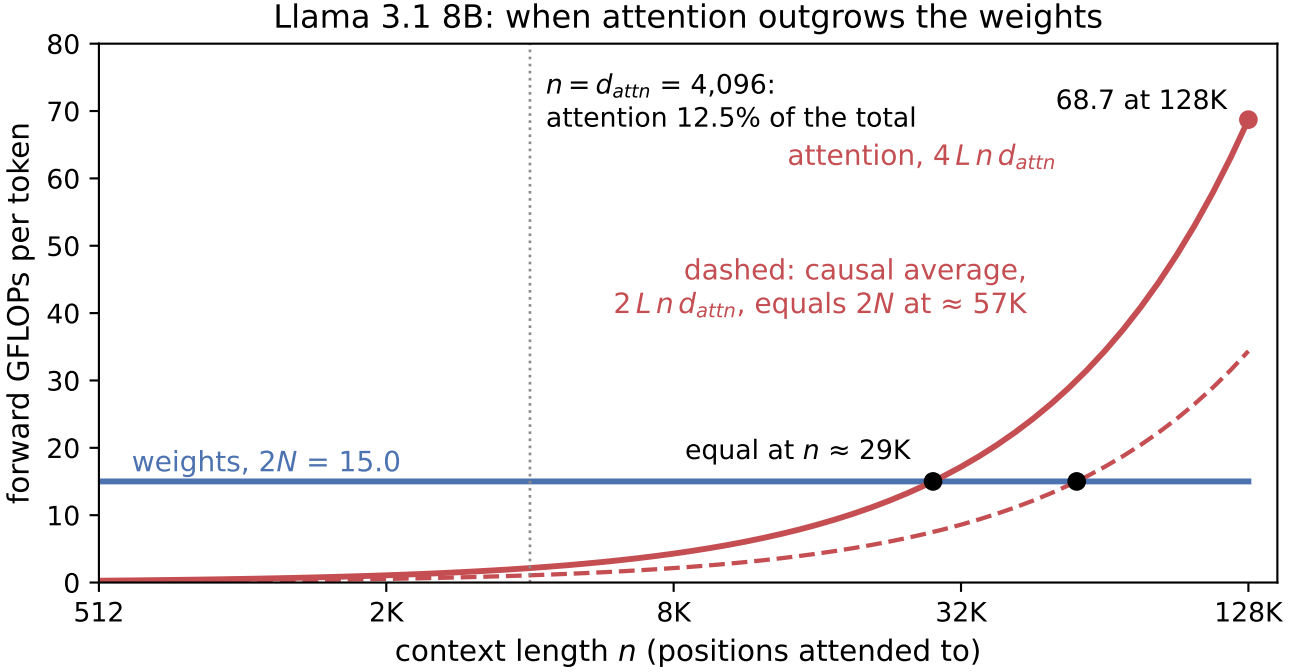
Llama 3.1 8B: when attention outgrows the weights
forward GFLOPs per token
context length n (positions attended to)
n = d_attn = 4,096:
attention 12.5% of the total
68.7 at 128K
attention, 4 L n d_attn
dashed: causal average,
2 L n d_attn, equals 2N at ≈ 57K
weights, 2N = 15.0
equal at n ≈ 29K
0
10
20
30
40
50
60
70
80
512
2K
8K
32K
128K